

MDC5011C • Responsible Technology Innovation Project

CYBERPRAHARI

National Cyber Threat Intelligence & Digital Ethics Platform

Forensic Analysis of 10 Authentic I4C Cyber Digest Incidents, Indian Statutory Penal Mapping (IT Act & BNS 2023), and Responsible Defensive Software Architecture

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VERIFIED REPOSITORIES:

github.com/Sujal-Shrivastav-07

github.com/Sujal-Shrivastav-07/ethics

 **Begin Presentation [Slide 02]** →

 **Jump to 10 I4C Cases**

 **Live Tool Demo**

 **Open GitHub Repository**

Presenter Note (30s): "Respected evaluators, I am Sujal Shrivastav. Today I present CyberPrahari, an innovative cyber threat intelligence platform grounded in authentic data from the Indian Cyber Crime Coordination Centre (I4C). We demonstrate how ethical design, statutory cyber laws, and defensive technology protect Indian citizens."

PROBLEM STATEMENT & THREAT LANDSCAPE

Systemic Failures in Indian Cyberspace

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1. The Semantic & Legal Literacy Deficit

Over 80% of consumer cyber fraud in India exploits **cognitive vulnerability and simulated state authority** rather than cryptographic hacking. Citizens fall prey to "Digital Arrest" because they are unaware that Indian law has no provision for video arrest.

2. Asymmetry in Incident Response (Golden Hour)

Financial cyber recovery depends entirely on the initial **120-minute "Golden Window"**. The majority of citizens are unaware of Helpline **1930** and the Citizen Financial Cyber Fraud Reporting System (CFCFRS) to freeze bank switches.

3. Structural Failures in Privacy by Design

Apps routinely demand blanket permissions (contact book and gallery scraping), state land registries expose unredacted citizen thumbprints online, and biometric payment scanners lack hardware liveness verification.

4. Weaponization of Dual-Use Technologies

Generative AI tools (real-time neural voice cloning, video face synthesis) and automated Telegram bot swarms operate without mandatory cryptographic provenance watermarks (C2PA) or legal transparency.

 [Jump to National Helpline 1930 Protocols →](#)

[Next: Project Scope \[Slide 03\] →](#)

Presenter Note (45s): "Our research revealed a crucial finding: modern cybercrime in India rarely attacks cryptographic algorithms. Instead, syndicates target human fear, greed, and institutional trust. From fake CBI video calls to cloned thumbprints from property deeds, attackers systematically exploit gaps in legal literacy and Privacy by Design."

PROJECT SCOPE & ACADEMIC OBJECTIVES

Objectives & Curricular Synthesis

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Core Project Objectives (100 Marks Rubric)

- **Authentic Research:** Extract and categorize 10 verified cyber incidents from official I4C Cyber Digest reports with zero data fabrication.
- **14-Parameter Forensic Dossiers:** Structure each incident across chronological attack chains, actor motivations, and security weaknesses.
- **Statutory Penal Mapping:** Map exact provisions across IT Act 2000, Bharatiya Nyaya Sanhita 2023 (BNS), and DPDP Act 2023.
- **Ethical Framework:** Formulate four pillars of digital ethics including Privacy by Design and developer codes of conduct.
- **Production Web Platform:** Build CyberPrahari with React 18, Vite, and Tailwind CSS with 100% zero-tracking local intelligence.

Curricular Alignment with All 10 Syllabus Titles

Our portfolio synthesizes all 10 project titles listed on Page 1 of the official guidelines:

- 1. I4C Cyber Digest Cybercrime & Ethics Portfolio
- 2. Cyber Threat Investigation using I4C Reports
- 3. Real-World Cybercrime Case Studies
- 4. Digital Ethics, Cyber Laws & Threat Intelligence
- 5. Responsible Tech Solutions for Cybercrime
- 6. Cyber Threat Intelligence Report (Recent Incidents)
- 7. Emerging Threats & Digital Governance Frameworks
- 8. Cybersecurity Awareness & Legal Analysis
- 9. Contemporary Threat Analysis (Daily Digest)
- 10. I4C Innovation Portfolio: Ethics, Laws & Practices

Next: Methodology [Slide 04] →

Direct to 10 Case Dossiers [Slide 05] →

Presenter Note (45s): "To ensure complete academic rigor, our project achieves five distinct goals: zero-fabrication research, 14-parameter forensic dossiers, legal mapping under BNS 2023, digital ethics frameworks, and an active web prototype synthesizing all ten recommended syllabus themes."

RESEARCH FRAMEWORK & METHODOLOGY

I4C Operational Framework & Case Selection

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Apex Intelligence Source

Indian Cyber Crime Coordination Centre (I4C) under the Ministry of Home Affairs (MHA), Gov. Operates the **National Cyber Crime Reporting Portal** (cybercrime.gov.in) and the **CFCFRS**.

Strict Selection Criteria

- 100% verified against advisories from I4C, [CERT-In](https://cert-in.org), UIDAI, and RBI.
- Diverse threat taxonomy (VoIP, Biometrics, Task scams, APK Trojans, Deepfakes, QR Quishing, Smishing).

14 Standardized Parameters

Each case evaluates: Incident Summary, Period, Source, Target, Attack Method, Crime Type, Attack Flow, Impact, Ethics, Laws, Weaknesses, Prevention, Responsible Tech, and Learnings.

Data Integrity Guarantee & Public Verification Portals

Zero data fabrication. Every case study directly maps to an official public alert issued by Gov (2023–2024).

cybercrime.gov.in

cert-in.org

[Explore 10 Cases →](#)

Presenter Note (45s): "Our research is grounded exclusively in authoritative public reports from I4C, CERT-In, UIDAI, and RBI. We selected ten distinct incident classes to ensure that our defensive models address the entire attack spectrum currently destabilizing Indian citizens."

CASE STUDIES & THREAT INTELLIGENCE

The 10 Authentic I4C Incident Cases

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#	Case Study Title	Primary Attack Vector	Targeted Demographic	Governing Statutes
01	"Digital Arrest" Extortion	Fake CBI/Customs video calls, psychological isolation	Senior citizens, retired judges	IT Sec 66D, BNS Sec 318, 204, 308
02	AePS Biometric Spoofing	Scraped deed prints molded into 3D silicone thumbs	Rural bank account holders	IT Sec 43/66, 66C, Aadhaar Act Sec 38
03	Telegram Task & "Like" Fraud	Prepaid Ponzi tasks via VIP Telegram bot groups	College youth, unemployed	IT Sec 66D, BNS Sec 318(4), PMLA 2002
04	Fake Institutional Stock Apps	Sideloaded rogue APKs showing fake 500% returns	Retail stock investors	IT Sec 66C/D, SEBI Act Sec 12A
05	Predatory Loan App Blackmail	Blanket permission harvesting, nude photo morphing	Financially stressed gig workers	IT Sec 66E, 67, BNS Sec 308, DPDP 2023
06	SIM Swap 2FA Hijacking	Forged telecom KYC to intercept banking SMS OTPs	High-net-worth individuals	IT Sec 43/66C, Telecom Act 2023
07	Electricity Disconnection Trojan	AnyDesk remote access screen mirroring via SMS	Residential utility consumers	IT Sec 43/66, BNS Sec 318
08	AI Deepfake Voice/Video Clone	30s neural voice clone & face swap in emergency calls	Parents of overseas students	IT Sec 66D, BNS Sec 318, 336
09	QR Code Quishing & Reverse UPI	"Scan QR to receive money" PIN debit deception	Online marketplace sellers	IT Sec 66D, BNS Sec 318(4), PSS Act
10	Courier Delivery Smishing APK	Fake postal SMS installing Trojan to steal 2FA OTPs	E-commerce shoppers	IT Sec 43/66, 66C, BNS Sec 318, 336

View 5-Stage Attack Chain [Slide 06] →

View Statutory Penal Mapping [Slide 08] →

Inspect in Live Web App

Presenter Note (60s): "These ten cases represent the definitive landscape of modern cyber fraud in India. Notice how each incident targets a specific vulnerability: Case 1 exploits fear of the state; Case 2 exploits unredacted biometrics; Case 5 weaponizes personal photo galleries; Case 8 weaponizes generative AI; Case 9 exploits UPI semantic confusion; and Case 10 weaponizes postal delivery trust."

FORENSIC THREAT MODELING

The Unified 5–Stage Cybercrime Attack Chain

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Universal 5–Stage Attack Lifecycle

- **Stage 1 (Reconnaissance):** Scrape public land registries, harvest social videos, broadcast bulk smishing SMS, spoof VoIP caller IDs.
- **Stage 2 (Psychological Hook):** Induce panic (arrest/power cutoff), greed (500% ROI), or urgency (parcel address missing).
- **Stage 3 (Technical Bypass):** Sideload malicious APKs, press silicone thumbs, capture screens via AnyDesk, reverse QR PIN traps.
- **Stage 4 (Asset Exfiltration):** High-velocity automated UPI, IMPS, and micro-ATM withdrawals routed to mule bank accounts.
- **Stage 5 (Laundering & Exit):** Rapid crypto conversion, international hawala routing, and communication severance.

Key Architectural Finding for Engineers

Attackers succeed because **technical exploitation is always preceded by psychological incapacitation**.

Defensive systems must intervene **before Stage 3**:

- Enforcing OS-level FLAG_SECURE to block remote screen mirroring.
- Hardware-level biometric liveness (blood pulse/thermal checks).
- Cognitive speed-bumps (cooling-off delays and vernacular warnings).

[Next: Responsible Tech Governance \[Slide 07\] →](#)[View Defensive Decision Support \[Slide 10\] →](#)

Presenter Note (45s): "Across all ten cases, we mapped a universal 5–stage attack chain. The critical takeaway for cybersecurity engineers is that technical defense must intervene before Stage 3—introducing cognitive speed-bumps and blocking remote mirroring before funds are transferred."

Responsible Technology Governance Framework

1. Privacy by Design

Mandate data minimization. No utility or lending application may demand contact lists or device galleries for service eligibility.

2. Consent Integrity

Ban dark patterns, forced bundled permissions, and pre-ticked checkboxes. Consent must be granular, explicit, and freely revocable.

3. Duty of Care

Platforms owe a moral duty to senior citizens and rural users, providing vernacular voice prompts and trusted-contact confirmations.

4. Anti-Deception & AI

Enforce mandatory cryptographic provenance watermarks ([C2PA standard](#)) on all generative AI audio/video synthesis engines.

Developer's Mandatory Code of Ethics

Engineers have a professional duty to refuse engineering deceptive interfaces, unvetted biometric capture, or permission-harvesting SDKs.

Next: Indian Cyber Laws [Slide 08]



Presenter Note (45s): "Ethics cannot be an afterthought. When a loan application demands access to your family photo album, or when public land portals expose citizen thumbprints online, that is a failure of digital ethics before it becomes a crime. Engineers must build Privacy by Design directly into architecture."

Indian Statutory Mapping (IT Act & BNS 2023)

Information Technology Act, 2000 (as amended)

- **Section 66D:** Cheating by personation using computer resource (Cases 1, 3, 4, 7, 8, 9, 10) — 3 yrs imprisonment + ₹1 Lakh fine.
- **Section 66C:** Identity theft (biometrics, passwords, tokens) (Cases 2, 4, 6, 8, 10) — 3 yrs imprisonment + ₹1 Lakh fine.
- **Section 66E:** Privacy violation (private area images) (Case 5) — 3 yrs imprisonment or ₹2 Lakhs fine.
- **Section 43 read with Sec 66:** Unauthorized data extraction, hacking & Trojans (Cases 2, 7, 9, 10) — 3 yrs or ₹5 Lakhs fine.

Bharatiya Nyaya Sanhita, 2023 (BNS) & DPDP Act

- **BNS Section 318(4):** Cheating and dishonestly inducing delivery of property (All 10 Cases; replaces IPC 420) — Up to 7 yrs rigorous imprisonment.
- **BNS Section 204:** Impersonating a public servant (Cases 1, 10; replaces IPC 170) — Up to 3 yrs imprisonment.
- **BNS Section 308:** Extortion by fear of injury (Cases 1, 5; replaces IPC 384) — Up to 3 yrs imprisonment.
- **DPDP Act 2023:** Non-consensual processing & data breaches — Penalties up to ₹250 Crores.
- **RBI Directives:** Customer Zero Liability in unauthorized electronic banking if reported within 3 days.

[IT Act 2000 \(India Code\)](#)[Bharatiya Nyaya Sanhita 2023](#)[DPDP Act 2023 \(MeitY\)](#)[Next: Technical Architecture \[Slide 09\] →](#)

Presenter Note (60s): "We have thoroughly mapped both the IT Act and the newly enacted Bharatiya Nyaya Sanhita, 2023. Notice Section 318(4) of BNS, which replaces the classic IPC Section 420 for cheating, and BNS Section 204 for impersonating police officers. Furthermore, the DPDP Act 2023 imposes massive fines on entities failing to protect biometric and personal data."

TECHNICAL ARCHITECTURE & STACK

The CyberPrahari Platform Architecture

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Frontend Stack

- **React 18** with functional hooks & state.
- **Vite Bundler** (sub-4s production build).
- **Tailwind CSS** (cybersecurity dark theme).
- **Lucide React Icons** for accessibility.

Local Deterministic Engine

- **Zero External Tracking** (no cookies/telemetry).
- **100% Client-Side Computation** for scam analysis and security audits.
- **Zero PII Input Required** (preserves user anonymity).

Containerized Deployment

- **Multi-stage Dockerfile** (Node build + Nginx Alpine web server).
- **Docker Compose** for single-command production deployment.

Core Application Modules & Live Access

Dashboard • 10-Case Dossiers • Scam Risk Analyzer • Cyber Hygiene Audit • Indian Law Matrix • 1930 Emergency Hub

 **Launch Local Prototype**

 **GitHub Repository**

Slide 10: Demo →

Presenter Note (45s): "To deliver a working solution, we engineered CyberPrahari. Built with React and Tailwind CSS, the platform operates entirely in the browser with zero data tracking, embodying the very Privacy by Design principles we advocate."

INTERACTIVE TOOLS DEMONSTRATION

Interactive Decision-Support Engines

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Tool 1: "Is This a Scam?" Risk Analyzer

- Evaluates incoming messages/calls against **10 weighted heuristic red flags**.
- Calculates real-time threat score (0–100) and outputs instant severity (Low, Moderate, High, Critical Danger).
- Delivers immediate concrete defensive protocols: *"Hang up immediately, refuse OTP disclosure, and dial 1930."*

Tool 2: Personal Cyber Hygiene Self-Audit

- Interactive 10–point actionable audit covering Aadhaar biometric locking, app-based 2FA, sideloading avoidance, and family safe words.
- Dynamic SVG completion ring & percentage score (0–100%).
- Tailored hardening recommendations based on identified user vulnerabilities.

 Open Live Tools in Web Browser

Next: 5 Golden Rules & 1930 [Slide 11] →

Back to 10 Cases [Slide 05]

Presenter Note (60s): "[Live Prototype Demo] Let me demonstrate our interactive tools live. If a citizen receives a call claiming their parcel was seized with contraband, ticking these flags in our Scam Analyzer immediately triggers 'CRITICAL SCAM DANGER', identifies Case 1 (Digital Arrest), and guides them to hang up and call 1930."

PUBLIC DEFENSE STRATEGY

5 Golden Rules & National Helpline 1930

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1. No Digital Arrest

Indian law enforcement NEVER conducts trials or issues arrests via Skype or WhatsApp video calls.

2. Lock Biometrics

Lock Aadhaar biometrics using mAadhaar app to neutralize AePS silicone thumb cloning.

3. Zero Unverified APKs

Never install APKs or remote tools (AnyDesk) sent over WhatsApp or SMS.

4. No Prepaid Tasks

Legitimate employment never requires capital deposits to release commissions.

5. Master the Golden Hour

Dial 1930 within 120 minutes to trigger automated banking freezes on CFCFRS.

Golden Rule of UPI

You NEVER enter a UPI PIN to RECEIVE money. Entering a PIN always debits your account.

[📞 Dial 1930 National Helpline](#)[🌐 cybercrime.gov.in Portal](#)[🏠 Sanchar Saathi Portal](#)[Next: Conclusion & Deliverables \[Slide 12\] →](#)

Presenter Note (45s): "Our awareness campaign distills these technical findings into the 5 Golden Rules of Citizen Defense. Most crucially, we emphasize the 2-Hour Golden Window; dialing 1930 triggers an automated freeze across Indian banking switches before stolen funds can be withdrawn at ATMs."

CONCLUSION & BIBLIOGRAPHY

Conclusion & Academic References

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Academic Deliverables Completed

- ✓ 10 Authentic I4C Incident Dossiers with 14 forensic parameters.
- ✓ Formal 13–Page Project Report (A4, Times New Roman, 1.5 line spacing).
- ✓ 12–Slide Seminar Presentation Deck in landscape PDF.
- ✓ Production Web Platform Prototype (React 18 + Dockerized).
- ✓ Printable Citizen Awareness Poster.
- ✓ Complete Statutory Mapping (IT Act, BNS 2023, DPDP Act).

Primary Academic References (IEEE Format)

- [1] I4C / MHA, "Advisory on Impersonation Frauds and Digital Arrest," 2024.
- [2] CERT-In, "Emerging Cyber Threats in Indian Cyberspace," Annual Report, 2023.
- [3] Ministry of Law & Justice, "The Information Technology Act, 2000 as amended."
- [4] Ministry of Law & Justice, "The Bharatiya Nyaya Sanhita, 2023 (Act 45 of 2023)."
- [5] MeitY, "The Digital Personal Data Protection Act, 2023 (Act 22 of 2023)."
- [6] UIDAI & RBI Security Directives on AePS & Digital Lending (2022–2023).
- [7] C2PA, "Technical Specification for Content Provenance and Authenticity," 2023.

 View 13–Page Project Report PDF

 View Awareness Poster PDF

 GitHub Repository

 Launch Web Prototype

Presenter Note (30s): "In conclusion, CyberPrahari bridges empirical threat intelligence, statutory cyber law, and responsible engineering. Thank you, respected evaluators. I am now ready for your questions."